

```
In [1]: # !pip install langchain
```

```
In [2]: # !pip install openai
```

```
In [3]: import os
        from dotenv import load_dotenv
        load_dotenv()
        OPENAI_API_KEY = os.getenv("OPENAI_API_KEY")
        HUGGINGFACEHUB_API_TOKEN = os.getenv("HUGGINGFACEHUB_API_TOKEN")
```

```
In [4]: from langchain_classic import PromptTemplate
        template = "{product}를 홍보하기 위한 좋은 문구를 추천해줘?"

        prompt = PromptTemplate(
            input_variables=["product"],
            template=template,
        )

        prompt.format(product="카메라")
```

```
Out[4]: '카메라를 홍보하기 위한 좋은 문구를 추천해줘?'
```

```
In [5]: from langchain_classic.chat_models import ChatOpenAI
        llm1 = ChatOpenAI(temperature=0,
                          model='gpt-4.1-mini'
                          )

        prompt = "진희는 강아지를 키우고 있습니다. 진희가 키우고 있는 동물은?"
        print(llm1.invoke(prompt).content)
```

C:\Users\User\AppData\Local\Temp\ipykernel_12876\3304050332.py:2: LangChainDeprecationWarning: The class `ChatOpenAI` was deprecated in LangChain 0.0.10 and will be removed in 1.0. An updated version of the class exists in the `langchain-openai` package and should be used instead. To use it run `pip install -U `langchain-openai` and import as `from `langchain_openai import ChatOpenAI``.

```
llm1 = ChatOpenAI(temperature=0,
진희가 키우고 있는 동물은 강아지입니다.
```

```
In [6]: llm2 = ChatOpenAI(
    base_url="https://router.huggingface.co/v1",
    api_key=HUGGINGFACEHUB_API_TOKEN,
    model="HuggingFaceTB/SmolLM3-3B",
    temperature=0,
)

prompt = "진희는 강아지를 키우고 있습니다. 진희가 키우고 있는 동물은?"
print(llm2.invoke(prompt).content)
```

<think>

Okay, the user is asking, "진희는 강아지를 키우고 있습니다. 진희가 키우고 있는 동물은?" which translates to "Hyunhee is raising a dog. What animal is Hyunhee raising?"

First, I need to confirm the information given. The sentence clearly states that Hyunhee is raising a dog. The question is essentially asking for the same information again. So the answer should be straightforward.

But maybe the user is testing if I can recognize the repetition or if there's a trick here. Let me check the original question again. The first part says "진희는 강아지를 키우고 있습니다." which is "Hyunhee is raising a dog." The second part is the same question phrased differently.

In Korean, sometimes questions can be structured in a way that's a bit confusing, but in this case, it's a direct question asking for the same information. The answer is clearly a dog.

I should make sure there's no ambiguity. The term "강아지" is unambiguously a dog in Korean. There's no other animal that's commonly referred to as "강아지" in this context.

So the answer is a dog. I don't see any need to complicate it further. The user might be practicing reading comprehension or testing if I can handle straightforward questions. Either way, the answer is clear.

</think>

진희가 키우고 있는 동물은 **강아지**입니다.

ModelLaboratory는 "교육용 실험 도구"라 보안을 전혀 고려 하지 않아서 compare로 출력하면 API_KEY가 다 출력되므로 사용하실 때는 주의!

```
In [7]: from langchain_classic.model_laboratory import ModelLaboratory
    model_lab = ModelLaboratory.from_llms([llm1, llm2])
    model_lab.compare("대한민국의 가을은 몇 월부터 몇 월까지야?")
```

Input:

대한민국의 가을은 몇 월부터 몇 월까지야?

```
client=<openai.resources.chat.completions.completions.Completions object at 0x000001601FD716A0> async_client=<openai.resources.chat.completions.completions.AsyncCompletions object at 0x000001601FD72120> model_name='gpt-4.1-mini' temperature=0.0 model_kwargs={} openai_api_key='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' openai_proxy=''
```

대한민국의 가을은 일반적으로 9월부터 11월까지입니다. 이 시기는 기온이 서서히 내려가고 단풍이 물드는 계절로, 가을 특유의 선선하고 쾌적한 날씨를 즐길 수 있는 때입니다.

```
client=<openai.resources.chat.completions.completions.Completions object at 0x000001601FE99450> async_client=<openai.resources.chat.completions.completions.AsyncCompletions object at 0x000001601FE99E50> model_name='HuggingFaceTB/SmolLM3-3B' temperature=0.0 model_kwargs={} openai_api_key='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' openai_api_base='https://router.huggingface.co/v1' openai_proxy=''
```

<think>

Okay, the user is asking about the months when autumn starts and ends in South Korea. Let me recall the traditional and modern definitions.

In traditional terms, the seasons are based on the solar terms, which are 24-hour periods. Autumn starts on the 8th solar term, which is "Liáng Qīng" (涼秋), and ends on the 22nd solar term, "Zhōng Shuǐ" (仲秋). These solar terms are calculated based on the sun's position in the sky, so they don't align with the Gregorian calendar months exactly.

However, in modern times, especially in South Korea, the seasons are often defined by the Gregorian calendar. Autumn is typically considered to start in September and end in November. But there might be some variation depending on the region or the source. For example, some sources might say September 1st to November 30th, while others might use the solar terms for more precise definitions.

I should also consider that in South Korea, the academic year starts in March, so the academic year's autumn semester might be from September to December. But the user is asking about the seasons, not the academic year.

So, to answer accurately, I should mention both the traditional solar terms and the modern Gregorian calendar months. It's important to note that the exact dates can vary slightly depending on the source, but the general consensus is that autumn in South Korea is from September to November.

</think>

대한민국의 가을은 **9월**부터 **11월**까지**으로** 정의됩니다.

정의:

- **가을 시작**: 9월 1일 (공식적으로) 또는 **8월 31일** (일정에 따라)
- **가을 종료**: 11월 30일 (공식적으로) 또는 **11월 30일** (일정에 따라)

참고 :

1. **기본적인 정의**:

- **가을**은 **9월**에서 **11월**까지의 **3개월**로 정의됩니다.
- **겨울**은 **12월**에서 **2월**까지의 **3개월**로 정의됩니다.

2. **기상학적 정의**:

- **가을**은 **9월**에서 **11월**까지의 기온이 **15℃** 이하로 떨어지면서 시작하며, **11월**에서 **12월**까지의 기온이 **0℃** 이하로 떨어지면서 끝납니다.

3. **기본적인 계절 분할**:

- **봄** : 3월 ~ 5월
- **여름** : 6월 ~ 8월
- **가을** : 9월 ~ 11월
- **겨울** : 12월 ~ 2월

추가 정보 :

- **기상청**의 공식적인 계절 분할은 **9월 1일**부터 **11월 30일**까지의 **가을**입니다.
- **일정에 따라** 일부 지역에서는 **8월 31일**부터 **11월 30일**까지의 **가을**을 고려할 수 있습니다.

따라서, **대한민국**의 가을은 **9월부터 11월까지**로 정의됩니다.

```
In [8]: from langchain_classic.output_parsers import CommaSeparatedListOutputParser
        from langchain_classic.prompts import PromptTemplate

        from langchain_classic.chat_models import ChatOpenAI
        llm = ChatOpenAI(temperature=0,
                        max_tokens=2048,
                        model_name='gpt-4.1-mini',
                        )

        output_parser = CommaSeparatedListOutputParser()
        format_instructions = output_parser.get_format_instructions()

        prompt = PromptTemplate(
            template="10개의 팀을 보여줘 {subject}.\n{format_instructions}",
            input_variables=["subject"],
            partial_variables={"format_instructions": format_instructions}
        )
```

```
In [9]: query = "한국의 야구팀은?"

# 출력 결과 생성
output = llm.invoke(prompt.format(subject=query)).content

# 출력에 대한 포맷 변경
parsed_result = output_parser.parse(output)
print(parsed_result)
```

```
['두산 베어스', 'LG 트윈스', '키움 히어로즈', '한화 이글스', '삼성 라이온즈', '롯데 자이언츠', 'KIA 타이거즈', 'NC 다이노스', 'S  
SG 랜더스', 'KT 위즈']
```